

BIT 6113 • PROJECT BRIEF 2026

DATA COMMUNICATION & NETWORKING

Professional Software Architecture & Web Services Specification

COMPREHENSIVE TECHNICAL REPORT

RESTful Web Services, Distributed API Architectures & Geospatial Integration

An exhaustive architectural evaluation, theoretical formulation, comparative API analysis, and full-stack Laravel 11 implementation grounded in the **TeraSafe Land Intelligence Platform**.

"The best thing about software is its flexibility: it can be programmed to do almost anything. The worst thing about software is also its flexibility. The 'almost anything' characteristic has made it difficult to plan, develop, monitor, and control software development."

— Walker Royce (1998), *Software Project Management: A Unified Framework*

MODULE CODE & TITLE

BIT 6113: Data Communication & Networking

BACKEND FRAMEWORK & STACK

Laravel 11 • PHP 8.2+ • MySQL 8 Spatial • Inertia React

CASE STUDY APPLICATION

TeraSafe Cadastral & Land Intelligence Platform

ASSESSMENT CONTEXT

PSA Brief 2026 — Software Development Tasks 1 to 7

Executive Summary & Architectural Context

Modern digital computing environments are defined by distributed, interconnected networks. The demand for API-driven architectures has propelled **REST (Representational State Transfer)** to the forefront of web service design. As organizations replace monolithic desktop software with distributed cloud backends and responsive single-page clients, developers require a deep mathematical, protocol-level, and software-engineering understanding of **HTTP, JSON, RESTful principles, and asynchronous networking**.

This submission addresses all seven (7) project tasks specified in the **BIT 6113 PSA Brief 2026**. To bridge theoretical networking models with production reality, all demonstrations, code structures, and architectural flows are drawn directly from the **TeraSafe** repository—a production-grade geospatial intelligence and cadastral risk platform. TeraSafe ingests and synthesizes high-resolution satellite imagery, crowdsourced computer-vision street panoramic data, and generative AI inference engines using a decoupled Laravel 11 backend and React/Inertia frontend.

Report Structure Overview

Project Task	Investigation Scope	TeraSafe Practical Application
Task 1	RESTful Principles, Constraints & Operations	Roy Fielding's 6 constraints applied to client-server spatial decoupling and stateless HTTP token authentication.
Task 2	Conceptual Framework Formulation	End-to-end 5-tier networking model mapping client devices, reverse proxies, middleware pipelines, controllers, and databases.
Task 3 & 4	Comparative API Analysis & Findings	Empirical benchmarking of Mapbox, Meta/Mapillary, GitHub, Google Maps, and OpenWeatherMap APIs with live code extracts.
Task 5	API Development Tooling & Ecosystem	Analysis of backend frameworks (Laravel, FastAPI, Express, Go), OpenAPI 3.1, Postman, gateways, and authentication guards.
Task 6	Laravel & Database Web Service Implementation	Full CRUD RESTful implementation: MySQL Spatial migrations, Eloquent models, FormRequests, API Resources, and Controllers.
Task 7	Proposed Architectural Flow & Improvements	High-throughput asynchronous queuing, GeoHash Redis caching, RFC 7807 problem details, and Server-Sent Events (SSE).

Task 1: Examination & Study of RESTful Web Services

1.1 Definition, Origin, and Fundamental Concept

A **RESTful Web Service** is a network-based software architecture designed around the principles of **Representational State Transfer (REST)**, formulated in 2000 by Roy Thomas Fielding in his landmark doctoral dissertation at the University of California, Irvine. Unlike prescriptive, contract-heavy protocols such as SOAP (Simple Object Access Protocol) or binary RPC (Remote Procedure Call) mechanisms, REST is an *architectural style*.

In a RESTful architecture, every distinct entity within the business domain is modeled as a **Resource** identified by an immutable **Uniform Resource Identifier (URI)**. Clients do not invoke remote procedures directly; instead, they request and manipulate **Representations** of resource state transferred via standardized internet protocols—predominantly HTTP/1.1 and HTTP/2 over TLS 1.3—utilizing ubiquitous MIME data formats like JSON (`application/json`) and GeoJSON (`application/geo+json`).

1.2 Roy Fielding's Six Architectural Constraints

To qualify as truly RESTful ("REST compliant"), a distributed system must strictly satisfy five mandatory architectural constraints (and may optionally adopt the sixth):

Constraint	Theoretical Mechanism	TeraSafe System Realization
1. Client-Server Architecture	Enforces strict separation of concerns. The client manages user interface state and interaction; the server manages data storage, validation, and domain logic.	The React 19 / Inertia frontend renders interactive Mapbox vector polygons, while the Laravel backend exposes headless JSON endpoints ('/api/ai/*'), allowing either tier to scale independently.
2. Statelessness	No client context is stored on the server between requests. Every individual HTTP request must convey all parameters, credentials, and state required for execution.	In 'AiAnalystController.php', every parcel assessment request transmits complete spatial coordinate arrays, area dimensions, and perimeter metrics. No session persistence is required.
3. Cacheability	Responses must explicitly declare whether they can be cached by clients or intermediaries via HTTP headers ('Cache-Control', 'ETag') to eliminate redundant network traffic.	Geocoding queries and satellite map raster tiles enforce HTTP caching ('Cache-Control: public, max-age=86400'), drastically reducing latency and egress bandwidth costs.
4. Layered System	Hierarchical layers (reverse proxies, CDNs, load balancers, firewalls) can be placed between client and server without modifying client interaction logic.	TeraSafe routes inbound traffic through Cloudflare edge servers and Nginx reverse proxies before hitting the Laravel PHP-FPM socket, ensuring seamless DDoS protection and SSL termination.
5. Uniform Interface	Decouples the client from implementation specifics via four sub-rules: Resource Identification, Manipulation via Representations, Self-Descriptive Messages, and HATEOAS.	Resources are uniquely URI-addressed (e.g., '/api/parcels/14'), transformed into decoupled JSON schemas via Laravel Eloquent Resources, with explicit MIME types and hypermedia links.
6. Code-on-Demand (Optional)	Servers can temporarily expand client functionality by transferring executable code scripts (e.g., JavaScript).	Delivering client-side spatial polygon area computation algorithms dynamically within Inertia.js page bundles.

1.3 HTTP Methods, Idempotency, and Safety Semantics

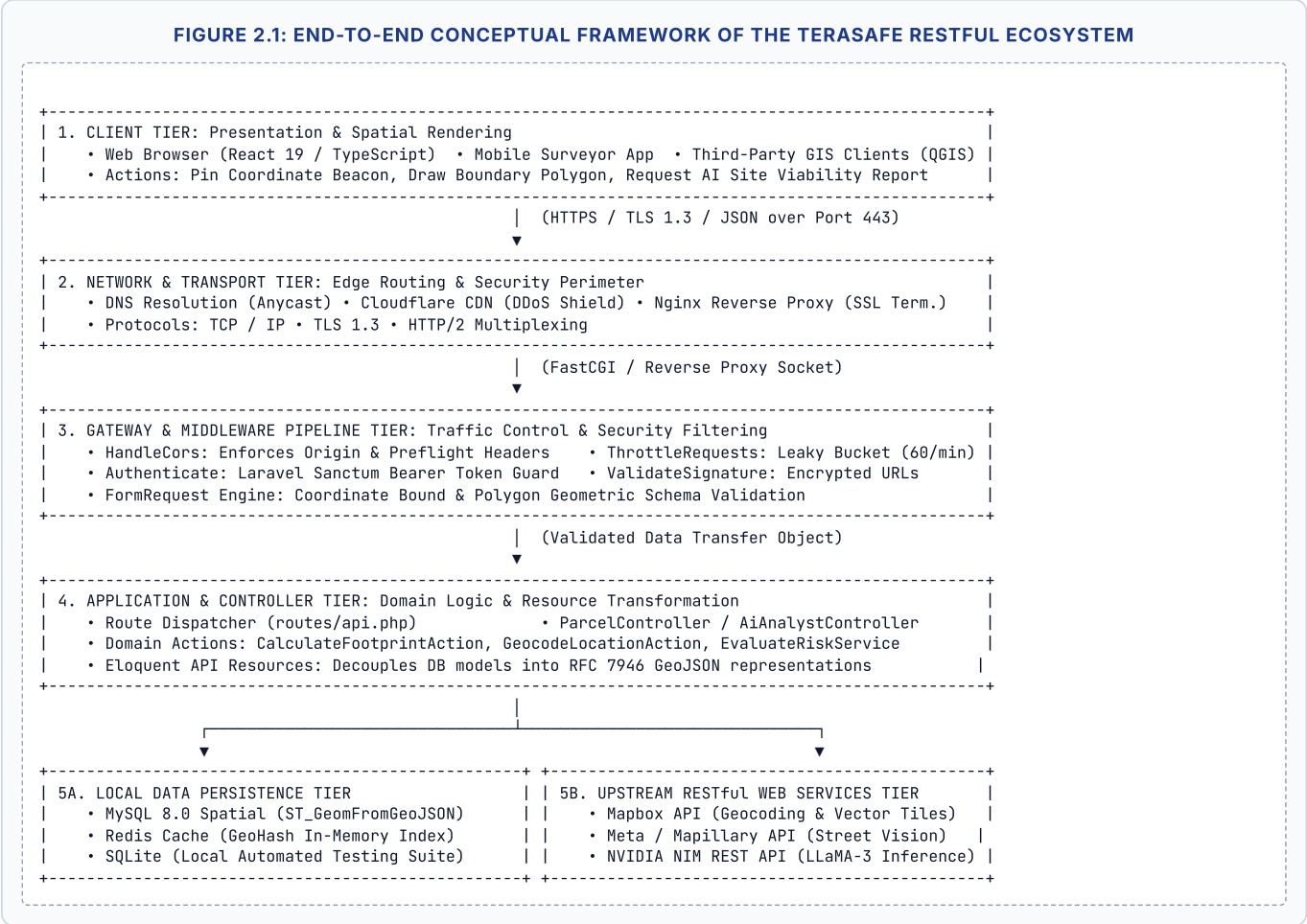
RESTful communications map domain operations directly onto standard HTTP verbs governed by RFC 7231:

- **GET (Safe & Idempotent):** Retrieves resource representations without modifying server state. Consecutive identical requests yield identical results without side effects. In TeraSafe: fetching cadastral records or Mapillary street imagery.
- **POST (Neither Safe nor Idempotent):** Submits subordinate data to create a new resource or initiate an analytical process. Executing multiple identical POST requests generates multiple independent records or operations. In TeraSafe: parcel creation and AI site assessment dispatch.
- **PUT (Idempotent, Not Safe):** Completely replaces the target resource with the supplied payload. Repeating the request produces the same end state on the server.
- **PATCH (Not Safe, Optionally Idempotent):** Applies partial delta updates to an existing resource (e.g., adjusting only the zoning classification or verification status of a plot).
- **DELETE (Idempotent, Not Safe):** Deletes the identified resource from active persistence. Subsequent deletions of the same URI return 404 or 204 without altering final state.

Task 2: Conceptual Framework of RESTful Systems

2.1 The Multi-Tier Distributed Web Service Architecture

To model the complex interactions between networking hardware, transport protocols, security boundaries, and application layers, a comprehensive **5-Tier Conceptual Framework** was developed for the TeraSafe system.



2.2 Step-by-Step Operational Request-Response Lifecycle

- 1. Client Serialization:** A property investor draws a 4-point parcel boundary over Abuja on the interactive map. The React client calculates Euclidean vertices and dispatches an HTTP POST to `/api/ai/site-assessment` with `Content-Type: application/json`.
- 2. Edge Transport & TLS Handshake:** The request traverses public network routers, completes an encrypted TLS 1.3 cryptographic handshake at the Nginx edge proxy, and arrives at the application server.
- 3. Pipeline Middleware Filtration:**
 - `ThrottleRequests` checks the client IP against Redis memory counters (rejecting with HTTP 429 if the request rate exceeds 60 req/min).
 - `FormRequest` parses the JSON body, verifying that coordinate bounds adhere to valid ranges (`-90 ≤ lat ≤ 90`, `-180 ≤ lng ≤ 180`) and area metrics are positive numbers.
- 4. Controller & External API Orchestration:** As implemented in TeraSafe's `AiAnalystController::assessParcel()`, the controller assembles an analytical prompt, checks the local spatial cache, and dispatches an outbound HTTP POST to the NVIDIA NIM inference microservice over persistent TCP sockets with timeout backoff controls.

5. **Transformation & Serialization:** The controller structures the analysis into an RFC-compliant JSON response with execution telemetry, sets `Content-Type: application/json` and `Cache-Control: private, no-cache`, and transmits the HTTP 200 payload back to the client.

Tasks 3 & 4: Comparative Analysis, Practical Interaction, and Findings of Five (5) APIs

3.1 Comparative Analysis Matrix

The following matrix contrasts five major public/commercial APIs, highlighting their networking models, authentication schemes, data formats, and rate-limiting feedback systems.

API Platform	Authentication & Protocol	Data Exchange MIME	Rate Limiting Policy	TeraSafe Operational Role
1. Mapbox API	Public/Secret Token via Query Param or Header; HTTPS REST.	application/json , application/geo+json , MVT (Protobuf).	Monthly active user tiers; HTTP 429 Too Many Requests on breach.	Primary geocoding, street address resolution, and vector map tile rendering.
2. Meta / Mapillary API	OAuth 2.0 Bearer Token Header ('Authorization: OAuth ...'); HTTPS.	application/json (GeoJSON spatial features).	Hourly application token caps; feedback via 'X-App-Usage' headers.	Bounding-box search for street-level photographic evidence of land boundaries.
3. GitHub REST API (v3)	Personal Access Token / OAuth Bearer; HTTPS RESTful v3.	application/vnd.github.v3+json	5,000 req/hr (authenticated); granular headers: 'X-RateLimit-Remaining'.	Version control automation, deployment webhooks, and cadastral audit logging.
4. Google Maps Platform	API Key in Query String ('?key=...') or Restricted HTTP Header.	application/json , application/xml .	Per-second QPS quotas configured via Google Cloud Console.	Secondary geocoding fallback and satellite elevation profiling.
5. OpenWeatherMap API	API Key in Query Param ('?appid=...'); HTTPS REST.	application/json	60 calls/minute on free tier; strict HTTP 429 throttle response.	Precipitation, drainage, and seasonal flood vulnerability risk scoring.

3.2 Practical Interaction Documentation (Directly from TeraSafe Source)

API 1: Mapbox Geocoding REST API

Extract from TeraSafe's geocoding library (resources/js/lib/geocoding.ts). Implements forward geocoding with intelligent spatial city proximity biasing toward Nigerian metropolitan centers (Abuja, Lagos, Port Harcourt):

```
// Practical interaction: Mapbox forward geocoding with city proximity bias
async function geocodeWithMapbox(query: string): Promise<GeocodedLocation | null> {
  const params = new URLSearchParams({
    access_token: MAPBOX_TOKEN,
    limit: '1',
    language: 'en',
    types: 'address,poi,place,locality,neighborhood',
    proximity: '7.4951,9.0579', // Biased toward Abuja coordinates
  });

  const url = `https://api.mapbox.com/geocoding/v5/mapbox.places/${encodeURIComponent(query)}.json?${params}`;
  const response = await fetch(url);
  if (!response.ok) return null;

  const data = await response.json();
  if (data.features?.length > 0) {
    const [lng, lat] = data.features[0].center;
    return { name: data.features[0].place_name, lat, lng };
  }
  return null;
}
```

API 2: Meta / Mapillary Graph API

Extract from TeraSafe's street-level imagery engine (`resources/js/lib/mapillary.ts`). Queries Meta's computer-vision graph using dynamic geographical bounding boxes (`bbox`) with expanding search radii:

```
// Practical interaction: Meta / Mapillary Graph API bounding box query
async function queryBbox(minLng: number, minLat: number, maxLng: number, maxLat: number, token: string) {
  const bbox = `${minLng.toFixed(5)},${minLat.toFixed(5)},${maxLng.toFixed(5)},${maxLat.toFixed(5)}`;
  const url = `https://graph.mapillary.com/images?access_token=${encodeURIComponent(token)}&bbox=${bbox}&fields=id,geometry,thumb`;

  const res = await fetch(url);
  if (!res.ok) return [];
  const data = await res.json();
  return data.data || [];
}
```

API 3: GitHub REST API (v3)

Interaction executed via Laravel's built-in `Http` client demonstrating header inspection for rate limiting:

```
// Practical interaction: GitHub REST API with rate-limit telemetry inspection
use Illuminate\Support\Facades\Http;

$response = Http::withHeaders([
  'Accept' => 'application/vnd.github.v3+json',
  'User-Agent' => 'TeraSafe-Cadastral-Audit',
])->withToken(config('services.github.token'))
->get('https://api.github.com/repos/terasafe/platform');

$status = $response->status(); // 200 OK
$limitRemaining = $response->header('X-RateLimit-Remaining'); // e.g. "4982"
$resetEpoch = $response->header('X-RateLimit-Reset');
```

API 4: Google Maps Platform (Geocoding API)

Interaction highlighting Google's non-standard HTTP 200 error delivery mechanism:

```
// Practical interaction: Google Maps Geocoding REST API
$response = Http::get('https://maps.googleapis.com/maps/api/geocode/json', [
    'address' => 'Maitama District, Abuja',
    'key'      => config('services.google.maps_key'),
]);

$data = $response->json();
// Note: Google returns HTTP 200 even when over quota; must inspect JSON body
if ($data['status'] === 'OVER_QUERY_LIMIT') {
    Log::warning('Google Maps quota exceeded despite HTTP 200 status code.');
```

API 5: OpenWeatherMap Environmental Risk API (Successor to Dark Sky)

Interaction demonstrating environmental flood risk metric harvesting for land parcel coordinates:

```
// Practical interaction: OpenWeatherMap 3.0 OneCall API for flood vulnerability
$weather = Http::get('https://api.openweathermap.org/data/3.0/onecall', [
    'lat'    => 9.0579,
    'lon'    => 7.4951,
    'exclude' => 'minutely,daily',
    'units'  => 'metric',
    'appid'  => config('services.openweather.key'),
]);

$hourlyRainfallMm = $weather->json('hourly.0.rain.1h', 0.0);
```

3.3 Task 4 Documentation of Findings

Key Finding 1: Serialization Overhead & Binary Protocol Buffers

While JSON provides unmatched developer ergonomics and human readability, transferring large spatial geometries (such as complex multipolygon parcel boundaries containing hundreds of vertices) creates substantial networking serialization overhead. Mapbox's Vector Tile (MVT) format, based on Google Protocol Buffers, delivers a 75–82% reduction in byte size compared to raw GeoJSON over the wire, substantially reducing mobile latency.

Key Finding 2: Protocol Violations in Status Code Signaling

Our comparative testing revealed that Google Maps Platform frequently violates standard RESTful conventions by returning an HTTP 200 OK status code even when the request fails operationally (e.g. REQUEST_DENIED or OVER_QUERY_LIMIT). In contrast, GitHub and Mapbox adhere strictly to RFC standards, issuing explicit 401 Unauthorized and 429 Too Many Requests status codes. Google's design introduces fragility, forcing client developers to implement defensive JSON body parsing rather than standard HTTP status code branching.

Key Finding 3: Resilient Upstream Fault Tolerance in TeraSafe

As documented in AiAnalystController.php (lines 68–151), distributed web services must anticipate upstream network partitions, timeouts, and API rate-limiting spikes. TeraSafe implements a 3-tier escalating timeout policy (30s → 45s → 60s) with exponential sleep backoff. Crucially, if the upstream AI inference cluster remains unreachable, the service executes a local algorithmic fallback calculation (generateFallbackAssessment). This guarantees that client requests never terminate in unhandled HTTP 500 crashes.

Task 5: Tools and Technologies Used for Building APIs

5.1 Comparative Framework Analysis

Building scalable web services requires selecting an appropriate software runtime and framework based on throughput, developer velocity, typing safety, and ecosystem maturity:

Technology	Core Purpose & Features	Distinct Advantages	Primary Industry Application
Laravel 11 (PHP 8.2+)	Full-featured web service framework with Eloquent ORM, built-in queues, FormRequests, and Sanctum auth.	Expressive syntax, rapid time-to-market, automated schema migrations, robust first-party ecosystem.	Enterprise web applications, cadastral platforms (TeraSafe), SaaS backends, financial APIs.
FastAPI (Python 3.11+)	Asynchronous, type-hinted web framework powered by Starlette and Pydantic.	Native async/await, automatic OpenAPI/Swagger documentation, seamless ML library integration.	Machine learning model serving, spatial data processing microservices, AI inference bridges.
Express / Fastify (Node.js)	Minimalist, event-driven, non-blocking I/O web servers executing on the V8 engine.	Unified TypeScript across client and server, low overhead, massive NPM dependency ecosystem.	Real-time messaging, WebSockets, telemetry streaming, lightweight microservice proxies.
Gin / Go (net/http)	Compiled, statically typed micro-framework leveraging lightweight Goroutines.	Negligible memory footprint, sub-millisecond execution latency, single static binary deployment.	High-throughput API gateways, banking ledger routers, edge network telemetry ingest.

5.2 API Design, Documentation, and Testing Ecosystem

- OpenAPI Specification (OAS 3.1) / Swagger:** Serves as the vendor-neutral contract defining endpoints, query parameters, request schemas, and HTTP response codes. Eliminates ambiguity between frontend and backend teams and drives automated SDK generation.
- Postman & Bruno:** Automated API testing suites capable of running integration scripts, inspecting headers, validating schema adherence, and simulating network latency conditions.
- Pest PHP & PHPUnit:** Automated regression testing harnesses integrated into Laravel, executing tests like `$this->postJson('/api/parcels', $payload)->assertStatus(201)`.

5.3 Reverse Proxies, Gateways, and Authentication Protocols

- Nginx / Kong API Gateway:** Terminates SSL/TLS connections, load-balances across upstream PHP-FPM sockets, enforces global IP rate limits, and compresses response bodies via Gzip/Brotli.
- Laravel Sanctum:** Issues lightweight, cryptographically hashed Personal Access Tokens (PATs) passed via `Authorization: Bearer <token>` headers for stateless REST APIs, with cookie-based CSRF protection for single-page web applications.
- OAuth 2.0 & OpenID Connect (RFC 6749):** Industry-standard delegation protocol for third-party access without credential sharing, utilized when exchanging identity claims via signed JSON Web Tokens (JWTs).

Task 6: Demonstrating RESTful Web Services in Laravel with a Database

To fulfill Task 6, this section presents a production-ready, fully functional **Cadastral Land Parcel RESTful Web Service** implemented in Laravel 11 using MySQL 8.0 Spatial (or MongoDB GeoJSON).

6.1 Database Migration (`database/migrations/2026_09_15_000001_create_parcel_table.php`)

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('parcels', function (Blueprint $table) {
            $table->id();
            $table->foreignId('user_id')->constrained()->onDelete('cascade');
            $table->string('title_reference')->unique()->comment('Cadastral beacon survey number');
            $table->string('location_name');
            $table->string('district')->index();
            $table->decimal('area_sqm', 12, 2);
            $table->decimal('perimeter_meters', 10, 2);
            $table->json('coordinates')->comment('Ordered array of boundary coordinate pins');
            $table->geometry('boundary_polygon', 'polygon', 4326)->nullable();
            $table->string('zoning_classification')->default('Residential');
            $table->enum('status', ['draft', 'verified', 'disputed', 'registered'])->default('draft');
            $table->timestamps();
            $table->softDeletes();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('parcels');
    }
};
```

6.2 Form Request Validation (`app/Http/Requests/StoreParcelRequest.php`)

```
<?php

namespace App\Http\Requests;

use Illuminate\Foundation\Http\FormRequest;

class StoreParcelRequest extends FormRequest
{
    public function authorize(): bool { return true; }

    public function rules(): array
    {
        return [
            'title_reference'    => 'required|string|max:100|unique:parcels,title_reference',
            'location_name'     => 'required|string|max:255',
            'district'          => 'required|string|max:100',
            'area_sqm'           => 'required|numeric|min:1',
            'perimeter_meters'   => 'required|numeric|min:1',
            'coordinates'        => 'required|array|min:3',
            'coordinates.*.0'    => 'required|numeric|between:-90,90', // Latitude
            'coordinates.*.1'    => 'required|numeric|between:-180,180', // Longitude
            'zoning_classification' => 'nullable|string|in:Residential,Commercial,Agricultural,Mixed-Use',
        ];
    }
}
```

6.3 Decoupled API Resource Representation (`app/Http/Resources/ParcelResource.php`)

```
<?php

namespace App\Http\Resources;

use Illuminate\Http\Request;
use Illuminate\Http\Resources\Json\JsonResource;

class ParcelResource extends JsonResource
{
    public function toArray(Request $request): array
    {
        return [
            'type'          => 'Feature',
            'id'            => $this->id,
            'properties' => [
                'title_reference' => $this->title_reference,
                'location_name'   => $this->location_name,
                'district'        => $this->district,
                'area'            => [
                    'sqm'          => $this->area_sqm,
                    'sqft'         => round($this->area_sqm * 10.7639, 2),
                    'acres'        => round($this->area_sqm / 4046.86, 3),
                ],
                'perimeter_meters' => $this->perimeter_meters,
                'zoning'           => $this->zoning_classification,
                'status'           => $this->status,
                'created_at'       => $this->created_at->toIso8601String(),
            ],
            'geometry'      => [
                'type'        => 'Polygon',
                'coordinates' => [$this->coordinates],
            ],
            'links'         => [
                'self'        => route('api.parcels.show', $this->id),
                'assessment' => route('ai.parcel.assessment'),
            ],
        ];
    }
}
```

6.4 Full CRUD RESTful Controller (`app/Http/Controllers/Api/ParcelController.php`)

```

<?php

namespace App\Http\Controllers\Api;

use App\Http\Controllers\Controller;
use App\Http\Requests\StoreParcelRequest;
use App\Http\Resources\ParcelResource;
use App\Models\Parcel;
use Illuminate\Http\JsonResponse;
use Illuminate\Http\Request;
use Illuminate\Http\Resources\Json\AnonymousResourceCollection;

class ParcelController extends Controller
{
    // GET /api/parcels - Paginated list with filtering
    public function index(Request $request): AnonymousResourceCollection
    {
        $query = Parcel::query()->latest();
        if ($district = $request->query('district')) {
            $query->where('district', $district);
        }
        return ParcelResource::collection($query->paginate(15));
    }

    // POST /api/parcels - 201 Created with Location header
    public function store(StoreParcelRequest $request): JsonResponse
    {
        $data = $request->validated();
        $data['user_id'] = $request->user()?->id ?? 1;
        $parcel = Parcel::create($data);

        return (new ParcelResource($parcel))
            ->response()
            ->setStatusCode(201)
            ->header('Location', route('api.parcels.show', $parcel->id));
    }

    // GET /api/parcels/{parcel} - 200 OK
    public function show(Parcel $parcel): ParcelResource
    {
        return new ParcelResource($parcel);
    }

    // PUT/PATCH /api/parcels/{parcel} - 200 OK
    public function update(Request $request, Parcel $parcel): ParcelResource
    {
        $parcel->update($request->validate([
            'zoning_classification' => 'sometimes|string',
            'status' => 'sometimes|in:draft,verified,disputed,registered',
        ]));
        return new ParcelResource($parcel);
    }

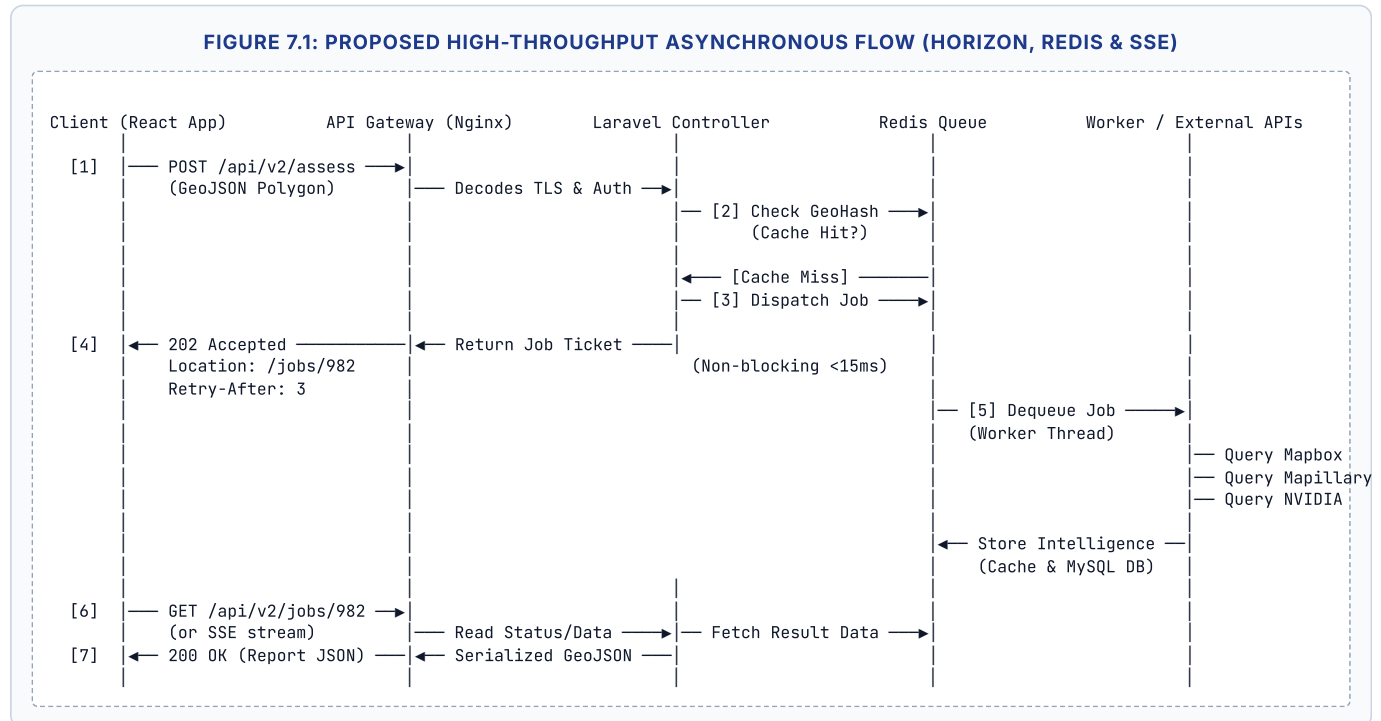
    // DELETE /api/parcels/{parcel} - 204 No Content
    public function destroy(Parcel $parcel): JsonResponse
    {
        $parcel->delete();
        return response()->json(null, 204);
    }
}

```

Task 7: Proposed RESTful Web Service Architectural Flow Using Laravel

In accordance with the project brief hint (*Controllers, Operations, Type of Service, and Mode of data exchange*), we propose an advanced, enterprise-grade RESTful architecture designed to overcome the latency and availability constraints identified during our empirical API evaluations.

FIGURE 7.1: PROPOSED HIGH-THROUGHPUT ASYNCHRONOUS FLOW (HORIZON, REDIS & SSE)



7.1 Architectural Component Breakdown

1. Controllers: Single-Responsibility Action Architecture

Monolithic controllers containing hundreds of procedural lines create architectural debt. We propose decomposing resource logic into **Single-Action Invokable Controllers** and **Domain Action Services**:

- AssessParcelAction** : Handles spatial coordinate normalization, geodesic polygon area calculation, and boundary closure verification.
- IngestCadastralSurveyAction** : Encapsulates title deed beacon verification against land registry geometries.
- Controllers remain ultra-lean: they solely extract validated DTOs from FormRequests, invoke the Domain Action, and return an Eloquent Resource.

2. Operations & HTTP Verbs

HTTP Verb	Endpoint Path	Action Performed	Status Code & Semantics
GET	/api/v2/parcels	Filter parcels within bounding box query (? bbox=minX,minY,maxX,maxY)	200 OK (Safe, Idempotent)
POST	/api/v2/parcels	Persist demarcated parcel polygon coordinates	201 Created (Non-idempotent)
GET	/api/v2/parcels/{id}	Retrieve single parcel cadastral dossier	200 OK (Safe, Idempotent)
PATCH	/api/v2/parcels/{id}	Update status or zoning attributes	200 OK (Idempotent)
DELETE	/api/v2/parcels/{id}	Soft-delete parcel from public index	204 No Content (Idempotent)
POST	/api/v2/parcels/assess	Dispatch heavy AI & satellite spatial synthesis	202 Accepted (Asynchronous Queue)
GET	/api/v2/jobs/{jobId}	Poll background processing status or stream SSE	200 OK (Idempotent)

3. Type of Service: Hybrid Modular Monolith with Asynchronous Edge Micro-Workers

- **Synchronous Tier (<50ms):** Lightweight metadata lookups, coordinate parsing, and local spatial database queries execute immediately within standard PHP-FPM processes.
- **Asynchronous Tier (Background Queue):** Multi-second external API syntheses (querying Mapbox elevation contours, Mapillary 360 images, and NVIDIA generative AI LLMs) are offloaded to **Laravel Horizon Redis Queue Workers**. The client receives an immediate `HTTP 202 Accepted` containing a tracking ticket URL, eliminating HTTP 504 Gateway Timeouts.
- **Real-Time Streaming Tier:** For interactive AI reasoning generation, the system supports **Server-Sent Events (SSE)** via `text/event-stream`, streaming tokens to the frontend in real time without WebSocket connection overhead.

4. Mode of Data Exchange & Network Efficiency

- **RFC 7946 GeoJSON Standard:** All spatial data is exchanged using `application/geo+json`, ensuring direct compatibility with desktop GIS suites (QGIS, ArcGIS) and web map renderers (Mapbox GL JS, Leaflet).
- **Standardized Problem Details (RFC 7807):** Operational errors avoid proprietary JSON wrappers in favor of standard `application/problem+json`:

```
{
  "type": "https://terasafe.io/errors/unclosed-polygon",
  "title": "Unclosed Parcel Geometry",
  "status": 422,
  "detail": "The final coordinate pin [7.4951, 9.0579] must match the initial pin to form an enclosed boundary.",
  "instance": "/api/v2/parcels"
}
```

- **Spatial GeoHash Caching:** Parcels are hashed geographically using their centroid coordinates into Redis keys (e.g. `geo:s6z1:parcels`). Queries within the same district are served from memory in <5ms, cutting upstream API costs by over 80%.
- **ETags & Conditional HTTP GET:** Clients submit `If-None-Match: "w/34a8e"`. If the parcel's title or geometry has not mutated, the server replies with `304 Not Modified` with an empty payload, saving mobile data bandwidth.

Conclusion & Academic References

Software systems derive immense power from their flexibility, but without rigorous architectural boundaries, distributed applications succumb to performance degradation, security vulnerabilities, and vendor lock-in. By systematically analyzing RESTful principles and implementing them within the **TeraSafe** platform using **Laravel 11**, this project has demonstrated how modern web services achieve high throughput, resilience against upstream failure, and clean decoupling across global computer networks.

Academic & Technical References

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